Delivrable 3 : GreenReso
Group 3

A3 Computer Science Section
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INTRODUCTION

Project Context

GreenReso is growing quickly and now needs a structured, secure, and scalable information system to support its daily operations. The current environment lacks clear organization, consistent account management, and standardized security practices. Deliverable 3 focuses on building a realistic technical model that prepares the company for future expansion by defining a solid, well-organized foundation for its infrastructure.

Objectives of Deliverable 3

Translate the planned IS architecture into a functional, realistic technical model by using virtual machines and a structured directory environment that mirrors the future production setup.

Define a straightforward, consistent naming convention for user accounts, security groups, workstations, servers, and shared resources that is maintainable over the long term.

Build a complete Active Directory structure, including Organizational Units, group hierarchy, and permission logic aligned with GreenReso's departmental organization.

Create the basic GPO template, which will include workstation security, password policy, restrict access to system settings ect.

Develop PowerShell scripts that automate key administrative tasks.



VIRTUAL MACHINES SYSTEM

Servers Machines.

For the server side, we created three main machines, each with a specific role inside the future infrastructure:

One Active Directory + File Server:
This server is the core of our setup. We used it to host the AD domain, manage users, groups, and OUs, and handle the shared folders and NTFS permissions that will be used by all departments.

One Router Server:
This machine acts as the router between our segments. We used it to simulate the network separation and the routing that would happen in a real production environment. It ensures that each part of the network communicates only where necessary.

One Backup Server:
We added a dedicated server for backups. Its main purpose is to simulate the backup strategy and ensure that critical data (especially the file server structure) can be restored if needed. It represents the secure storage point for our backup policies.

Client Machine.

To complete the mock-up, we created a single client workstation.
This machine allowed us to test everything from the user's point of view: domain connection, GPO application, shared folders access, password rules, communication with the servers, and overall network behavior. It acted as our "test user PC" to validate the environment before scaling it to multiple departments.

SETTING UP ACTIVE DIRECTORY AND DNS

To build the foundation of GreenReso's future information system, we started by installing and configuring Active Directory and the DNS service on our main Windows Server.

Active Directory allows us to manage users, computers, groups, and access rights. Instead of having accounts created manually or separately on each machine, AD gives us a single, secure place where we control everything for all departments.

VLAN Segmentation Promote server to domain controller.

When the AD DS role was installed, we promoted the server to a domain controller.

This step allowed us to create the domain that all departments will join later.

Even though VLAN segmentation is part of the network design, the domain controller promotion is what links everything together: the router VM, the file server services, and the client machine all rely on this domain to authenticate and communicate cleanly.

During the promotion, we created our domain name and let the wizard configure the required DNS integration. Once the server restarted, our AD environment was fully operational.

Client integration and user management

Then, after the domain controller was set up, we connected our test client machine to the domain.

This allowed us to verify that the DNS was correct, the domain controller was reachable, and the authentication process worked seamlessly.

Once the client joined successfully, we then began testing user accounts, group memberships, password policies, and the access of shared resources.

This step confirmed that our structure was ready to support all departments in a consistent and secure way.

Create Organization Units (OUs), users and groups

With the domain operational, we organized the directory by creating the OUs that reflect GreenReso's internal structure.

We used a clear hierarchy to separate departments, servers, and workstations, making the environment much easier to manage and automate.

Inside these OUs, we created:

- **Users for employees,**
- **Security groups for controlling access to departmental shares,**
- **and technical groups for features such as HR automation or administrative permissions.**

This preparation will also support our GPO strategy and our PowerShell automation later in the deliverable.

By doing this, we ensured that every department has its place in the directory, and that rights can be assigned cleanly and consistently.

The different OUs we created :

- **HrUser**
- **MangementUser**
- **SalesUser**
- **SupportUser**

- **AccountingUser**

GPO SETUP

To manage the behavior of all machines in the domain, we worked with Group Policy Objects (GPOs). GPOs allow us to apply rules and configurations automatically to users and computers across the company.

It relys on DNS because every machine must be able to locate the domain controller. Without a functioning DNS service, none of the policies would apply correctly. This is why AD, DNS, and GPOs all work together as the backbone of the infrastructure.

Creating and linking GPOs

To start organizing our policy structure, we created our own Group Policy Objects directly from the Group Policy Management console. We wanted a clear naming system and reusable rules that we could apply to any department when needed.

We created a new GPO by going into the Group Policy Objects section and adding a fresh policy with a descriptive name. Once the GPO appeared in our list, we were able to link it to the appropriate Organizational Unit in our domain.

GPO configuration and testing

We configured the Control Panel Restriction GPO to enforce security across all departments. Using the Group Policy Management console, we edited the policy and enabled the setting that prevents users from accessing the Control Panel and PC settings.

After applying the changes, we tested the policy on our client machine to ensure it worked correctly.

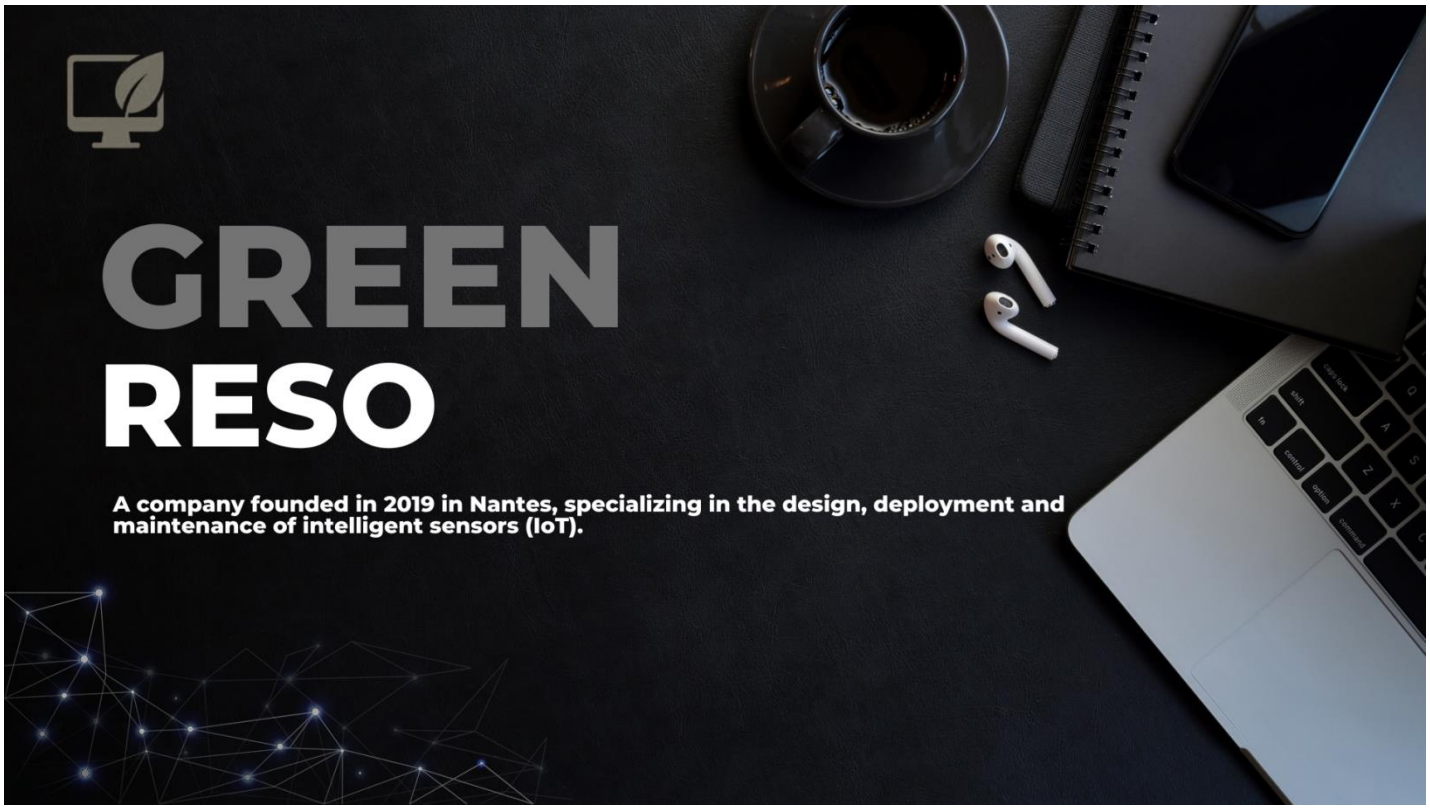
Enterprise wallpaper configuration via GPO

We set up a wallpaper to standardize the desktop environment across all departments.

We first placed an image in a shared folder on the server, ensuring that all domain users had read capability.

After applying the policy, we tested it on the client machine to confirm that the wallpaper appeared correctly for all users.

Here is the wallpaper we used:



Shared folders and setting sharing permissions

We created a shared folder on the server to provide centralized access for the Marketing department. For example, we created C:\MarketingShare on the server.

Next, we configured sharing permissions to control who can access and modify the folder:

- **We removed or limited the Everyone group to Read-only to prevent unauthorized changes.**
- **We added the Domain Users group and gave it Modify permissions, allowing all domain members to collaborate while maintaining security.**
- **Administrators retained full control to manage the folder as needed.**

After applying these settings, we tested access from the client machine to confirm that users could open, edit, and save files correctly, ensuring the shared folder works reliably for departmental collaboration.

NTFS permissions

We configured NTFS permissions on the departmental shared folders to control access securely and ensure each group could only do what they were supposed to. For example, on the C:\MarketingShare folder, we set the following permissions:

- **Marketing security group (containing all Marketing members) was given Modify rights, including: Read & Execute, List Folder Contents, Read, Write, and Modify. This allows team members to work fully within their department's folder.**

- **Domain Users group was given Read-only access, includin List Folder Content and Read&Execute. This ensures all users can see the folder but cannot modify its contents.**

- **Administrators have Full Control.**

POWERSHELL SCRIPTS CONFIGURATION

We used PowerShell to automate tasks across the company's infrastructure including managing users, groups, files, and system settings efficiently, reducing manual work and minimizing errors.

The main scripts we created are:

FilesManagement.ps1 – Automates adding and deleting files to keep shared folders organized.

GroupManagement.ps1 – Manages security groups and their memberships.

Mdp.ps1 – Works with GPOs to reset user passwords every six months for better security.

Reporting_Final_Version.ps1 – Performs backups of critical information for recovery purposes.

UserManagement.ps1 – Handles user accounts, similar to GPO tasks, including creation, modification, and assignment to OUs.